

YANG, YAO

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1. SUMMARY

Completed MSc in Artificial Intelligence at HKU and BEng in Data Science and Big Data Technology at NEU, YANG, YAO combines a solid theoretical grounding in mathematics and statistics with deep practical experience in algorithm programming. His research interests focus on (1) (Multimodal) Large Language Models, (2) Agents, and (3) Safe and Trustworthy AI, while staying OPEN to exploration across broader AI topics.

2. EDUCATION

MSc in Artificial Intelligence, University of Hong Kong **Sep 2023 - Jun 2025**
GPA of 3.24/4.00, Faculty of Science **Hong Kong SAR**

► Core courses: Optimization Theory, Machine Learning, Computational Intelligence, and Deep Learning

BEng in Data Science and Big Data Technology, Northeastern University **Sep 2019 - Jun 2023**
GPA of 3.76/5.00 (87.6/100.0), School of Mathematics and Statistics **Shenyang & Qinhuangdao**

► Core courses: Advanced Algebra, Mathematical Analysis, Probability Theory, Bayes Statistics, Random Process, Python Programming, Distributed Clusters, MySQL Database, Operating System, and Computer Networks

3. INTERNSHIPS

Zhuang Liu Lab, Princeton University **May 2026 - Present**
Agents Research Intern **Princeton**

Safety and Trustworthiness AI Center, Shanghai Artificial Intelligence Laboratory **Nov 2024 - Present**
Large Language Models Research Intern / Research Assistant **Shanghai**

4. PUBLICATIONS & SUBMISSIONS

research has started since early 2025

- [1] **Yang Yao***, Haozhe Chen*, Bingyi Kang, Karthik R Narasimhan, and Zhuang Liu. 2026. SWEeper-Bench: Can Agents Discover Bugs in Interactive Software? (Under Review)
- [2] **Yang Yao**, Xuan Tong, Ruofan Wang, et al. 2025. A Mousetrap: Fooling Large Reasoning Models for Jailbreak with Chain of Iterative Chaos. (ACL 2025)
- [3] **Yang Yao**, Lingyu Li, Jiaxin Song, et al. 2025. Argus Inspection: Do Multimodal Large Language Models Possess the Eye of Panoptes? (ACM Multimedia 2025, Oral)
- [4] **Yang Yao (Core Contributor)**, et al., and Bowen Zhou. 2025. SafeWork-R1: Coevolving Safety and Intelligence under the AI-45° Law. (Intern SafeWork Technical Report)
- [5] **Yang Yao**, Yixu Wang, Yuxuan Zhang, et al. 2026. Dr. Bench: A Multidimensional Evaluation for Deep Research Agents, from Answers to Reports. (Under Review, **Received All Positive Scores (5/4/4) at ICML 2026** while regrettably rejected by PC)
- [6] Yixu Wang, **Yang Yao**, Xin Wang, et al. 2026. Towards Context-Invariant Safety Alignment for Large Language Models. (ICML 2026)
- [7] Lingyu Li, **Yang Yao**, Yixu Wang, et al. 2025. The Other Mind: How Language Models Exhibit Human Temporal Cognition. (AAAI 2026)
- [8] Yixu Wang, Xin Wang, **Yang Yao**, et al. 2025. AgenticEval: Toward Agentic and Self-Evolving Safety Evaluation

of Large Language Models. (ACL 2026)

- [9] Ruofan Wang, Xin Wang, **Yang Yao**, et al. 2026. Simulated Ensemble Attack: Transferring Jailbreaks Across Fine-tuned Vision-Language Models. (Under Review)
- [10] Chiyu Chen, Xinhao Song, Yunkai Chai, **Yang Yao**, et al. 2025. GhostEI-Bench: Do Mobile Agent Resilience to Environmental Injection in Dynamic On-Device Environments? (ICLR 2026)
- [11] Yixu Wang, Jiaxin Song, Yifeng Gao, Xin Wang, **Yang Yao**, et al. 2025. SafeVid: Toward Safety Aligned Video Large Multimodal Models. (NeurIPS 2025)
- [12] Xin Wang, Yunhao Chen, Juncheng Li, Yixu Wang, **Yang Yao**, et al., and Xia Hu. 2026. OpenRT: An Open-Source Red Teaming Framework for Multimodal LLMs. (OpenRT Technical Report)
- [13] Tianle Gu, et al., **Yang Yao**, et al. 2025. Probing the Safety Robustness of LLMs in Latent Space. (ACL 2026)
- [14] Xuan Tong, et al., **Yang Yao**, et al. 2025. CADiff: Context-Aware Diffusion for Controllable Anomaly Generation in Anomaly Detection. (AAAI 2026)

5. ACADEMIC EXPERIENCES

beyond the above publications and submissions

Multimodal Value Alignment: Benchmark and Reward Model Fine-tuning Nov 2024 - Mar 2025

- Constructed benchmarks with Multimodal Large Language Models (MLLMs, e.g., GPT, Gemini, and Qwen) based on the scenarios and elements of real-world human ethical values to evaluate the responses of MLLMs in terms of value alignment on response quality, security, reliability, and compliance.
- Built and optimized two pipelines based on image generation with DALL·E 3 and search with Google Image Search, and generated about 200,000 multimodal value data layer by layer through prompts engineering.
- Designed Outcome Reward Model (ORM) by full and Low-Rank Adaptation (LoRA) Supervised Fine-Tuning (SFT) Qwen2.5-VLs (7B, and 72B) and Process Reward Model (PRM), with accuracy rates of 93.9% and 96.8%.

Stochastic Optimization Method for AUC Maximization Aug - Oct 2024

- Explored the optimization methods for the Area Under receiver operating characteristic Curve (AUC) maximization based on the equivalent objectives, focusing on the fact that the classical stochastic optimization methods are not directly applicable to AUC maximization due to the nonlinearity of the objective functions.
- Designed diverse forms of alternative loss functions based on mathematical tricks (e.g., property of moment generating function, and Taylor expansion), and verified the convergence and stability of the algorithms.
- Achieved AUC scores of 0.9296 and 0.7197 respectively on the simple and difficult benchmarks of imbalanced classification tasks, surpassing the baseline.

6. PROFESSIONAL SKILLS

Python (PyTorch, TensorFlow, VeRL, LLMs, Agents, NumPy, Pandas, Matplotlib, Scikits-Learn), R, MySQL, Apache Hadoop Distributed Clusters (MapReduce & Zookeeper & Hive & HBase & Spark & Flume & Kafka), etc.

7. LEADERSHIP & TEAMWORK EXPERIENCES

Student Assistant Principal, Northeastern University at Qinhuangdao Dec 2020 - Dec 2021
Coach of Debate Team, Sydney Smart Technology College, Northeastern University Sep 2020 - Feb 2021

8. QUALIFICATIONS & PRIZES & COMPETITIONS

University Scholarships (4 times) & Innovation and Entrepreneurship Scholarships (2 times) Undergraduate
Honourable Mention, Mathematical and Interdisciplinary Contest in Modeling (MCM-ICM) May 2022
Third Prize, Chinese Mathematics Competitions (CMC) Dec 2020